DataCite Corpus - Data Exploration & Descriptive Statistics

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DataCite Corpus - Exploratory Analysis

We first look at the number and proportions of values in the DataCite corpus that are missing values for relevant variables: publication DOI/identifier, dataset creation date, article publication date, and funder information. We also return the total number of observations in the DataCite Corpus and the number of observations where the NIH is listed as a funder.

```
#### FUNCTION: get_dc_summary - argument filename (string) ####
## Argument filename takes the name of a DataCite Corpus
## file. Column names are assumed to match the columns
## provided in the DataCite Corpus as of 11 October 2026.
## Documentation and file downloads available
## at https://zenodo.org/records/16901115

get_dc_summary <- function(filename) {
  ## Returns descriptive statistics of the data file - number
  ## of rows, number & proportion of rows with publication DOIs,
  ## number & proportion of rows with a 'created' date,
  ## number & proportion of rows with publication dates,
  ## number & proportion of rows with a funder listed.

  dat <- read_csv(file = filename, na = c("", "NA", "none"))

  summarylist <- dat |> summarise(
    n = nrow(dat),
    n_doi = str_detect(dat$publication, pattern = "doi") |> sum(na.rm = TRUE),
    n_created_date = sum(1 - is.na(dat$created), na.rm = TRUE),
    n_pub_date = sum(1 - is.na(dat$publishedDate), na.rm = TRUE),
    n_funders = sum(1 - is.na(dat$funders), na.rm = TRUE),
    n_nih = str_detect(dat$funders, pattern = "(NIH)|(National Institutes of Health)") |>
      sum(na.rm = TRUE)
  ) |> unlist()

  return(summarylist)
}

#### FUNCTION: summary_datacite - argument filenames (string or vector of strings) ####
## Argument filenames stores names of DataCite Corpus files.
## Calls  get_dc_summary to compute summary statistics for
## each data file, returns a dataset.
summary_datacite <- function(filenames) {
  cl <- makeCluster(8)
  registerDoParallel(cl)
```

```
  summ_dc <- foreach(
    fn = filenames,
    .combine = "rbind",
    .export = "get_dc_summary",
    .packages = "tidyverse"
  ) %dopar% {
    get_dc_summary(fn)
  }

  stopCluster(cl)

  return(summ_dc)
}

#### Call summary_datacite ####

# Call summary_datacite, get dataset of summary statistics for
# each data file, reset row names to indexes
summary_dat <- summary_datacite(files) |> as.data.frame()
row.names(summary_dat) <- c(1:nrow(summary_dat))

# Compute summary statistics across summary_dat (for whole dataset)
summary_dat |> summarise(
  N.obs = sum(n),
  N.doi = sum(n_doi),
  Prop.doi = sum(n_doi) / sum(n),
  N.cr.date = sum(n_created_date),
  Prop.cr.date = sum(n_created_date) / sum(n),
  N.funders = sum(n_funders),
  Prop.funders = sum(n_funders) / sum(n),
  N.NIH = sum(n_nih),
  Prop.NIH = sum(n_nih) / sum(n)
) |>
  t() |>
  kable(digits = 4)
```

N.obs	10697745.0000
N.doi	10696293.0000
Prop.doi	0.9999
N.cr.date	10697745.0000
Prop.cr.date	1.0000
N.funders	97200.0000
Prop.funders	0.0091
N.NIH	5202.0000
Prop.NIH	0.0005

NIH-Funded Datasets

We then look at the number of datasets listed in the DataCite Corpus that list NIH (U.S. National Institutes of Health) as a funder. We check both the funders and fundersROR columns.

```r
#### FUNCTION: nih_subset - argument filename (string) ####
## Argument filename takes the name of a DataCite Corpus file.
## The function returns the subset of observations that list NIH
## as a funder.
get_nih_subset <- function(filename) {
  dat <- read_csv(file = filename, na = c("", "NA", "none")) |>
    filter(
      if_any(any_of(c("funders", "fundersROR")),
             ~str_detect(., pattern = "(NIH)|(National Institutes of Health)"))
    )
  return(dat)
}

#### FUNCTION: nih_subset - argument filenames (string or vector of strings) ####
## Argument filenames stores names of  DataCite Corpus files. Calls
## get_nih_subset to return the subset of all DataCite data files,
## that list NIH as a funder. Returns a dataset.
nih_subset <- function(filenames) {
  cl <- makeCluster(8)
  registerDoParallel(cl)

  summ_dc <- foreach(
    fn = filenames,
    .combine = "rbind",
    .export = "get_nih_subset",
    .packages = "tidyverse"
  ) %dopar% {
    get_nih_subset(fn)
  }

  stopCluster(cl)

  return(summ_dc)
}

# Get subset of DataCite Corpus that are NIH-funded
nih_dat <- nih_subset(files) |> as.data.frame()

# Compute summary statistics across for NIH-funded datasets only
nih_dat |> summarise(
  N.obs = nrow(nih_dat),
  N.doi = sum(!is.na(publication)),
  Prop.doi = sum(!is.na(publication)) / nrow(nih_dat),
  N.cr.date = sum(!is.na(created)),
  Prop.cr.date = sum(!is.na(created)) / nrow(nih_dat),
) |>
  t() |>
  kable(digits = 4)
```

N.obs	5291
N.doi	5291
Prop.doi	1
N.cr.date	5291

Retraction Watch Database

```
# load retraction watch data
rw <- data.frame(read.csv(
  "https://raw.githubusercontent.com/ashnryn/article-retractions/refs/heads/main/retraction-watch/retrac
  na.strings = c("", "unavailable")
))

#### Retraction Watch entries without DOIs ####
rw <- rw |> filter(is.na(OriginalPaperDOI))

# What journals are represented?
rw |> count(Journal) |> arrange(desc(n)) |> head(20)
```

```
##                                                                                                  Journa
## 1                2010 International Conference on Computer Application and System Modeling (ICCASM 201
## 2                                2010 2nd International Conference on Computer Engineering and Technolog
## 3                                        2010 International Conference on E-Business and E-Governmer
## 4                          2010 IEEE International Conference on Advanced Management Science (ICAMS 201
## 5                         2010 3rd International Conference on Computer Science and Information Technolog
## 6                2010 The 2nd Conference on Environmental Science and Information Application Technolog
## 7                              2010 2nd International Conference on Mechanical and Electronics Engineerir
## 8                                  2010 2nd IITA International Conference on Geoscience and Remote Sensir
## 9                                          2010 International Conference on Management and Service Scien
## 10                    2011 International Conference on Consumer Electronics, Communications and Networl
## 11                                 2010 2nd International Conference on Industrial and Information Syster
## 12                       2010 2nd IEEE International Conference on Information Management and Engineerir
## 13                       2010 International Conference on Optics, Photonics and Energy Engineering (OPE
## 14 2010 International Conference on Computer and Communication Technologies in Agriculture Engineerir
## 15                         2010 4th International Conference on Bioinformatics and Biomedical Engineerir
## 16                                    2011 International Conference on E-Business and E-Government (ICE
## 17                                                          JBUON: Journal of Balkan Union of Oncolog
## 18                                        International Journal of Clinical and Experimental Patholog
## 19                         2010 International Conference on Educational and Information Technology (ICEI
## 20                         2010 2nd International Conference on E-business and Information System Securi
##     n
## 1  282
## 2  254
## 3  216
## 4  208
## 5  198
## 6  138
## 7   84
## 8   81
## 9   80
## 10  57
## 11  55
## 12  51
## 13  49
## 14  48
## 15  44
```

```
## 16  40
## 17  40
## 18  38
## 19  34
## 20  33
```

```
rw |> filter(str_detect(Journal, pattern = "Conference")) |> nrow() # Majority are conference papers
```

```
## [1] 2290
```

```
# Which countries are represented?
rw |> count(Country) |>
  mutate(p = round(n / nrow(rw), 4)) |>
  arrange(desc(n)) |>
  head(20) # Majority were published in China
```

```
##                Country    n      p
## 1                China 2306 0.6231
## 2        United States  178 0.0481
## 3                India  107 0.0289
## 4          South Korea   86 0.0232
## 5                 Iran   85 0.0230
## 6              Unknown   81 0.0219
## 7              Romania   76 0.0205
## 8                Japan   49 0.0132
## 9              Germany   42 0.0113
## 10      United Kingdom   40 0.0108
## 11              Taiwan   35 0.0095
## 12              Canada   33 0.0089
## 13            Pakistan   27 0.0073
## 14              Poland   27 0.0073
## 15            Thailand   25 0.0068
## 16            Malaysia   24 0.0065
## 17              Turkey   24 0.0065
## 18           Singapore   23 0.0062
## 19               Italy   21 0.0057
## 20 China;United States   18 0.0049
```

```
rw <- rw |>
  mutate(across(c(OriginalPaperDate, RetractionDate), as.character)) |>
  mutate(pub.year = mdy_hm(OriginalPaperDate) |> year() |> as.integer(),
         .after = OriginalPaperDate) |>
  mutate(ret.year = mdy_hm(RetractionDate) |> year() |> as.integer(),
         .after = RetractionDate)

rw$pub.year[which(is.na(rw$pub.year))] = 1753

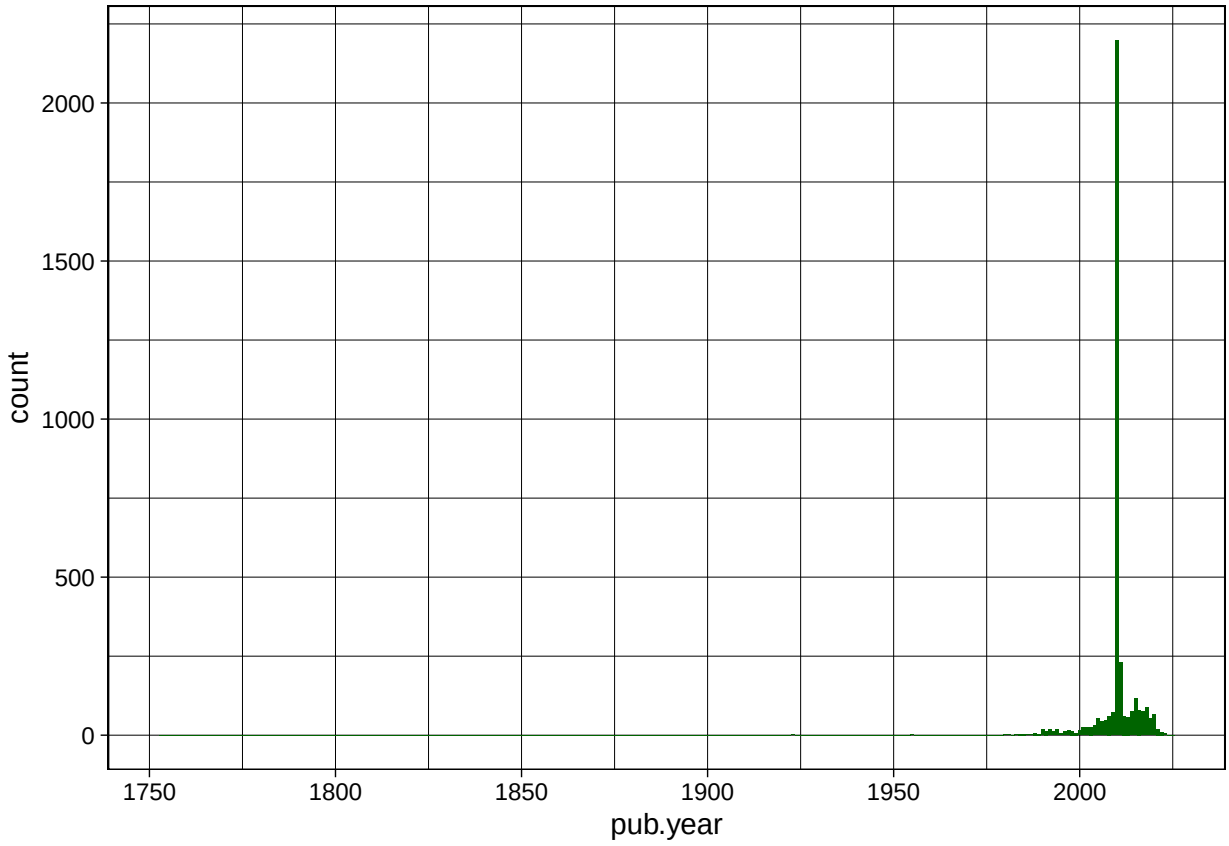
# Min and max publication dates?
cbind(min.date = min(rw$OriginalPaperDate),
      max.date = max(rw$OriginalPaperDate))
```

```
##      min.date               max.date
## [1,] "1/1/1753 12:00:00 AM" "9/9/2016 0:00"
```

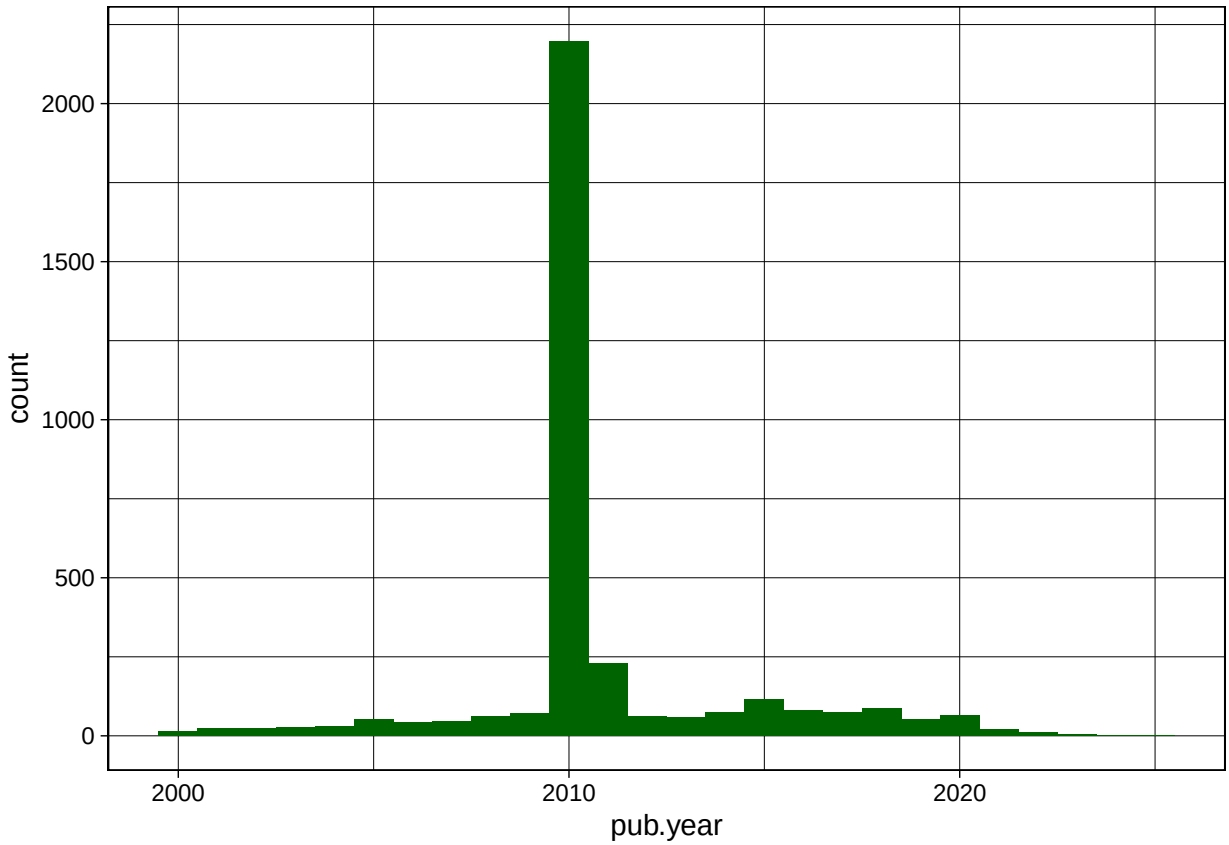
```
## The 1953 one is the "first ever english language retraction", per RW
## Benjamin Wilson, "Treatise upon Electricity"
```

```
## 2016 is "Expression of FPN and Hepcidin in breast cancer tissues and
## association of their expression with occurrence of anemia", Shihui Ma,
## Feihai Ling, et al.

# Retractions without DOIs, by publication year
rw |>
  ggplot(aes(x = pub.year)) +
    geom_histogram(binwidth = 1, fill = "darkgreen") +
    theme_linedraw()
```




```
# Restrict to 2000-2026
rw |> filter(!is.na(pub.year)) |>
  filter(pub.year >= 2000 & pub.year <= 2026) |>
  ggplot(aes(x = pub.year)) +
    geom_histogram(binwidth = 1, fill = "darkgreen") +
    theme_linedraw()
```




```
# Majority were published in 2010 specifically??
rw |> count(pub.year) |> arrange(desc(n))
```

```
##    pub.year    n
## 1      2010 2198
## 2      2011  229
## 3      2015  116
## 4      2018   88
## 5      2016   80
## 6      2017   75
## 7      2014   74
## 8      2009   72
## 9      2020   66
## 10     2008   61
## 11     2012   61
## 12     2013   57
## 13     2005   53
## 14     2019   53
## 15     2007   46
## 16     2006   44
## 17     2004   30
## 18     2003   26
## 19     2002   25
## 20     2001   24
## 21     2021   20
## 22     1992   18
## 23     1990   17
```

```
## 24     1994   17
## 25     1997   15
## 26     2000   15
## 27     1991   13
## 28     1998   13
## 29     1996   12
## 30     1993   11
## 31     2022   10
## 32     1988    7
## 33     1999    7
## 34     1995    6
## 35     2023    6
## 36     1984    4
## 37     1985    4
## 38     1989    4
## 39     1983    3
## 40     1987    3
## 41     1923    2
## 42     1955    2
## 43     1980    2
## 44     1981    2
## 45     1986    2
## 46     1753    1
## 47     1926    1
## 48     1951    1
## 49     1971    1
## 50     1978    1
## 51     1982    1
## 52     2024    1
## 53     2025    1
```

DataCite Corpus merged with Retraction Watch

```
df <- read_csv("merged.csv")

df |> summarise(
  N.obs = nrow(df),
  N.doi = sum(!is.na(data.citation.id)),
  Prop.doi = sum(!is.na(data.citation.id)) / nrow(df),
  N.cr.date = sum(!is.na(data.created.date)),
  Prop.cr.date = sum(!is.na(data.created.date)) / nrow(df),
  N.data.dois = sum(str_detect(data.id, pattern = "10\\.")),
  Prop.data.dois = sum(str_detect(data.id, pattern = "10\\.")) / nrow(df),
  N.pub.date = sum(!is.na(pub.date)),
  N.funders = sum(!is.na(pub.date)) / nrow(df),
  Prop.funders = sum(!is.na(data.funders)) / nrow(df),
  N.NIH = sum(str_detect(data.funders, pattern = "(NIH)|(NATIONAL INSTITUTES OF HEALTH)") |
                str_detect(data.funders.ror, pattern = "(NIH)|(NATIONAL INSTITUTES OF HEALTH)"),
              na.rm = TRUE),
  Prop.NIH = sum(str_detect(data.funders, pattern = "(NIH)|(NATIONAL INSTITUTES OF HEALTH)") |
                str_detect(data.funders.ror, pattern = "(NIH)|(NATIONAL INSTITUTES OF HEALTH)"),
                na.rm = TRUE) / nrow(df)
) |>
```

```
  t() |>
  kable(digits = 4)
```

N.obs	14970.0000
N.doi	14970.0000
Prop.doi	1.0000
N.cr.date	14970.0000
Prop.cr.date	1.0000
N.data.dois	1274.0000
Prop.data.dois	0.0851
N.pub.date	14970.0000
N.funders	1.0000
Prop.funders	0.0010
N.NIH	4.0000
Prop.NIH	0.0003

```
df |>
  filter(
      str_detect(data.funders, pattern = "(NIH)|(NATIONAL INSTITUTES OF HEALTH)") |
                str_detect(data.funders.ror, pattern = "(NIH)|(NATIONAL INSTITUTES OF HEALTH)")
    ) |>
  select(data.title, data.repository, pub.title, pub.retraction.nature, pub.retraction.reason)
```

```
## # A tibble: 4 x 5
##   data.title                     data.repository pub.title pub.retraction.nature
##   <chr>                          <chr>           <chr>     <chr>
## 1 PITTSBURGH YOUTH STUDY OLDEST~ ICPSR - INTERU~ Chronic ~ Correction
## 2 PITTSBURGH YOUTH STUDY DRUG C~ ICPSR - INTERU~ Chronic ~ Correction
## 3 PITTSBURGH YOUTH STUDY MENTAL~ ICPSR - INTERU~ Chronic ~ Correction
## 4 PITTSBURGH YOUTH STUDY DEMOGR~ ICPSR - INTERU~ Chronic ~ Correction
## # i 1 more variable: pub.retraction.reason <chr>
```

There's one retracted paper associated with a data in the DataCite Corpus that also appears in Retraction Watch. Also one of the papers we talked about in the previous article!

Number of repositories

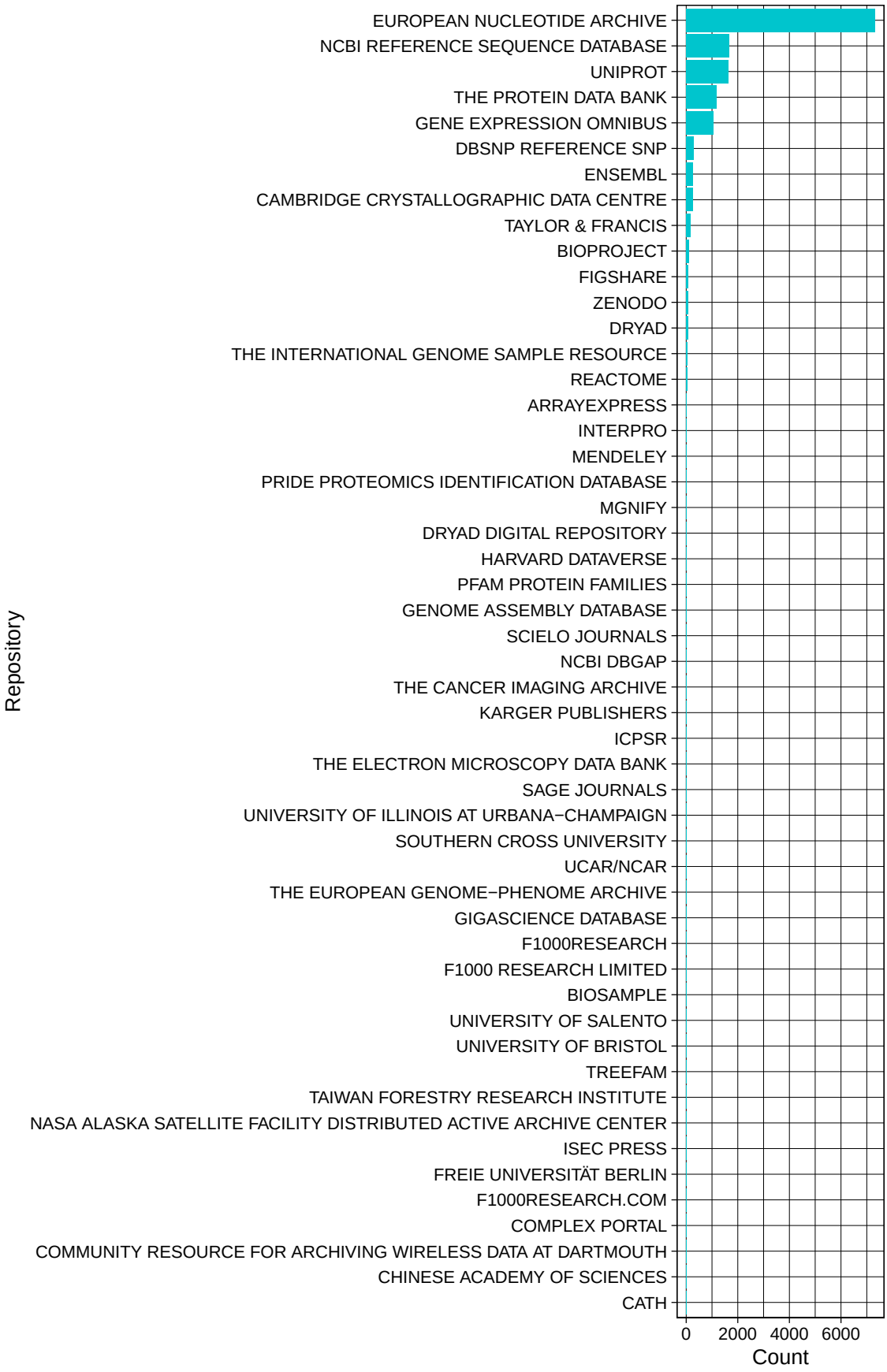
See which repositories are associated with retracted papers (tables and data visualizations).

```
## For all datasets, with OR without DOIs
# Table
df |> count(data.repository) |> arrange(desc(n))
```

```
## # A tibble: 53 x 2
##    data.repository                          n
##    <chr>                                <int>
##  1 EUROPEAN NUCLEOTIDE ARCHIVE           7316
##  2 NCBI REFERENCE SEQUENCE DATABASE      1655
##  3 UNIPROT                               1641
##  4 THE PROTEIN DATA BANK                 1178
##  5 GENE EXPRESSION OMNIBUS (GEO)         1052
##  6 <NA>                                   519
##  7 DBSNP REFERENCE SNP                    294
##  8 ENSEMBL                                273
```

```
##  9 CAMBRIDGE CRYSTALLOGRAPHIC DATA CENTRE   265
## 10 TAYLOR & FRANCIS                         177
## # i 43 more rows
```

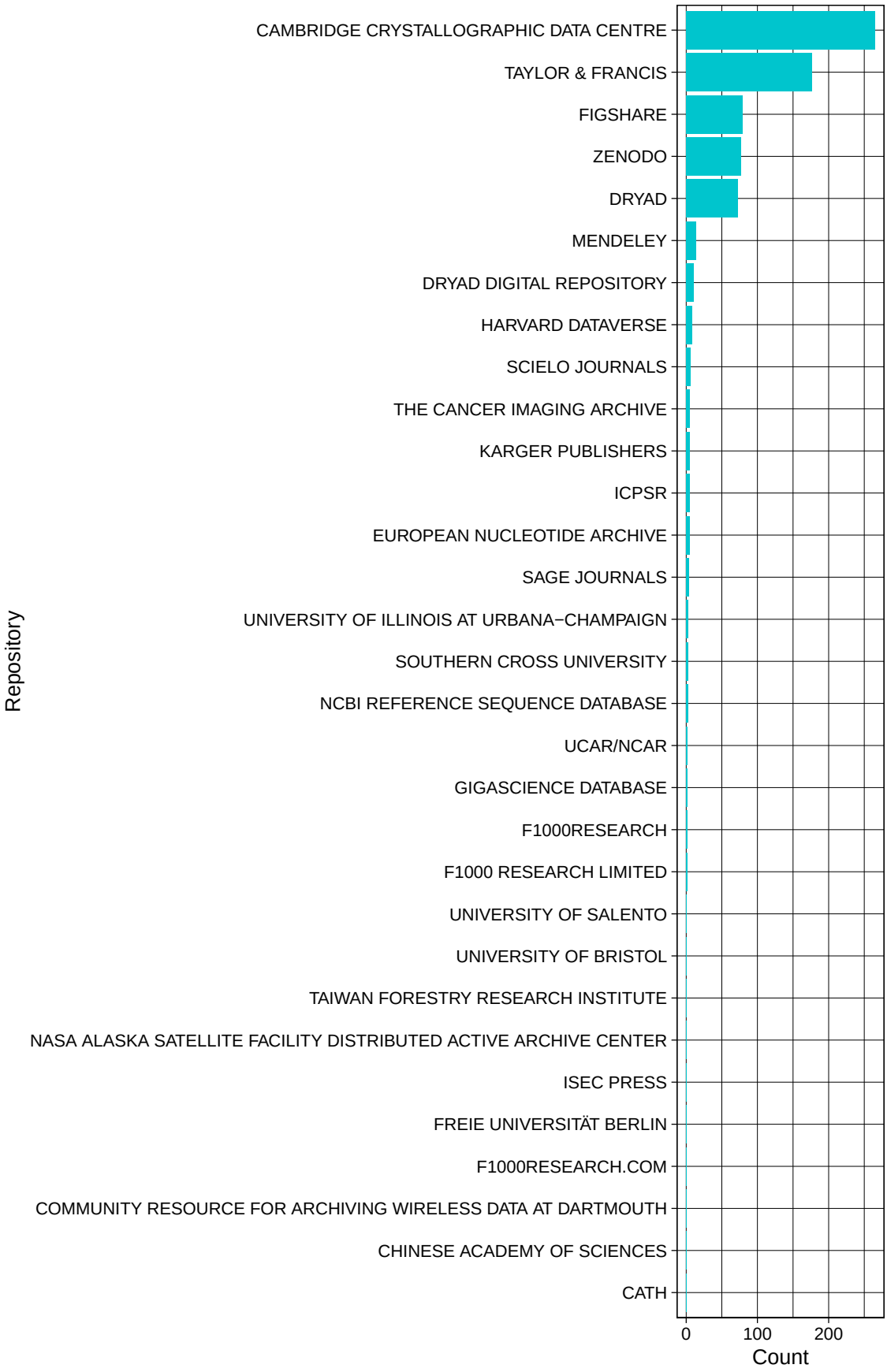
```r
# Bar graph
df |>
  filter(!is.na(data.repository) & data.repository != "UNPUBLISHED") |>
  mutate(
      data.repository.mod = str_remove(data.repository, pattern = "\\s\\(.*\\)")
    ) |>
  mutate(
      data.repository.mod = str_remove(data.repository.mod, pattern = "\\s-\\s.*")
    ) |>
  count(data.repository.mod) |>
  ggplot(aes(x = reorder(data.repository.mod, n), y = n)) +
    geom_bar(stat = "identity", fill = "turquoise3") +
    labs(x = "Repository", y = "Count") +
    coord_flip() +
    theme_linedraw()
```


EUROPEAN NUCLEOTIDE ARCHIVE
NCBI REFERENCE SEQUENCE DATABASE
UNIPROT
THE PROTEIN DATA BANK
GENE EXPRESSION OMNIBUS
DBSNP REFERENCE SNP
ENSEMBL
CAMBRIDGE CRYSTALLOGRAPHIC DATA CENTRE
TAYLOR & FRANCIS
BIOPROJECT
FIGSHARE
ZENODO
DRYAD
THE INTERNATIONAL GENOME SAMPLE RESOURCE
REACTOME
ARRAYEXPRESS
INTERPRO
MENDELEY
PRIDE PROTEOMICS IDENTIFICATION DATABASE
MGNIFY
DRYAD DIGITAL REPOSITORY
HARVARD DATAVERSE
PFAM PROTEIN FAMILIES
GENOME ASSEMBLY DATABASE
SCIELO JOURNALS
NCBI DBGAP
THE CANCER IMAGING ARCHIVE
KARGER PUBLISHERS
ICPSR
THE ELECTRON MICROSCOPY DATA BANK
SAGE JOURNALS
UNIVERSITY OF ILLINOIS AT URBANA−CHAMPAIGN
SOUTHERN CROSS UNIVERSITY
UCAR/NCAR
THE EUROPEAN GENOME−PHENOME ARCHIVE
GIGASCIENCE DATABASE
F1000RESEARCH
F1000 RESEARCH LIMITED
BIOSAMPLE
UNIVERSITY OF SALENTO
UNIVERSITY OF BRISTOL
TREEFAM
TAIWAN FORESTRY RESEARCH INSTITUTE
NASA ALASKA SATELLITE FACILITY DISTRIBUTED ACTIVE ARCHIVE CENTER
ISEC PRESS
FREIE UNIVERSITÄT BERLIN
F1000RESEARCH.COM
COMPLEX PORTAL
COMMUNITY RESOURCE FOR ARCHIVING WIRELESS DATA AT DARTMOUTH
CHINESE ACADEMY OF SCIENCES
CATH
Repository
0
2000
4000
6000
Count

```r
## For datasets without DOIs
# Table
df |> filter(str_detect(data.id, pattern = "10\\.")) |>
  count(data.repository) |> arrange(desc(n))
```

```
## # A tibble: 33 x 2
##    data.repository                              n
##    <chr>                                    <int>
##  1 <NA>                                       512
##  2 CAMBRIDGE CRYSTALLOGRAPHIC DATA CENTRE     265
##  3 TAYLOR & FRANCIS                           177
##  4 FIGSHARE                                    79
##  5 ZENODO                                      77
##  6 DRYAD                                       73
##  7 MENDELEY                                    14
##  8 DRYAD DIGITAL REPOSITORY                    10
##  9 HARVARD DATAVERSE                            8
## 10 SCIELO JOURNALS                              6
## # i 23 more rows
```

```r
# Bar graph
df |> filter(str_detect(data.id, pattern = "10\\.")) |>
  filter(!is.na(data.repository) & data.repository != "UNPUBLISHED") |>
  mutate(
      data.repository.mod = str_remove(data.repository, pattern = "\\s\\(.*\\)")
    ) |>
  mutate(
      data.repository.mod = str_remove(data.repository.mod, pattern = "\\s-\\s.*")
    ) |>
  count(data.repository.mod) |>
  ggplot(aes(x = reorder(data.repository.mod, n), y = n)) +
    geom_bar(stat = "identity", fill = "turquoise3") +
    labs(x = "Repository", y = "Count") +
    coord_flip() +
    theme_linedraw()
```


CAMBRIDGE CRYSTALLOGRAPHIC DATA CENTRE
TAYLOR & FRANCIS
FIGSHARE
ZENODO
DRYAD
MENDELEY
DRYAD DIGITAL REPOSITORY
HARVARD DATAVERSE
SCIELO JOURNALS
THE CANCER IMAGING ARCHIVE
KARGER PUBLISHERS
ICPSR
EUROPEAN NUCLEOTIDE ARCHIVE
SAGE JOURNALS
UNIVERSITY OF ILLINOIS AT URBANA−CHAMPAIGN
SOUTHERN CROSS UNIVERSITY
NCBI REFERENCE SEQUENCE DATABASE
UCAR/NCAR
GIGASCIENCE DATABASE
F1000RESEARCH
F1000 RESEARCH LIMITED
UNIVERSITY OF SALENTO
UNIVERSITY OF BRISTOL
TAIWAN FORESTRY RESEARCH INSTITUTE
NASA ALASKA SATELLITE FACILITY DISTRIBUTED ACTIVE ARCHIVE CENTER
ISEC PRESS
FREIE UNIVERSITÄT BERLIN
F1000RESEARCH.COM
COMMUNITY RESOURCE FOR ARCHIVING WIRELESS DATA AT DARTMOUTH
CHINESE ACADEMY OF SCIENCES
CATH
Repository
0
100
200
Count